you're interested in learning more about this subject, check out our Robot Creation Course.

What exactly is ROS?

Robot Operating System (ROS) is an acronym for Robot Operating System. Even though the name implies otherwise, ROS is not a true operating system because it is built on top of Ubuntu Linux (also on top of Mac, and recently, on top of Windows). ROS is an operating system framework that abstracts the hardware from the software. This implies you may consider all of the robot's hardware in terms of software. And this is great news for you since it means you can write robot programmes without having to worry about the hardware. Yea!

ROS is compatible with Ubuntu and Debian Linux. Although there is experimental support for OSX and Gentoo, and a version for Windows is in the works, we don't recommend using these unless you are an expert. More information on how to use ROS on certain systems can be found on this page.

ROS for service robots

At least in the field of service robots, ROS is quickly becoming the industry standard for robotics programming. ROS began at universities, but it quickly spread throughout the business world. Every day, more and more enterprises and startups are relying on ROS to run their operations. Every robot had to be programmed using the manufacturer's API before ROS. This meant that if you switched robots, you'd have to reinstall the entire software, in addition to learning the new API. In order to comprehend how your software was working, you also needed to know a lot about how to interact with the robot's electronics. It was comparable to the situation with computers in the 1980s when each computer had its own operating system and you had to write the same programme for each. ROS is the operating system for robots, similar to how Windows is for computers